Pairs of Continuous R.V.

1. X, Y - pair of continuous R.V., having the joint PDF:

$$f_{X,Y}(x,y) = \begin{cases} c, & x+y \geq 1,\ x \geq 0,\ y \geq 0 \\ \text{zero, otherwise} \end{cases}$$

a. $c = ?$ $c \in \mathbb{R}$

b. $P(\underline{X \leq Y}) = ?$ $\quad A = \{X \leq Y\}$

c. $P(\underline{X + Y \leq \frac{1}{2}}) = ?$ $\quad B = \{X + Y \leq \frac{1}{2}\}$

d. $f_{X,Y|A}(x,y) = ?$ $\quad f_{X,Y|B}(x,y) = ?$

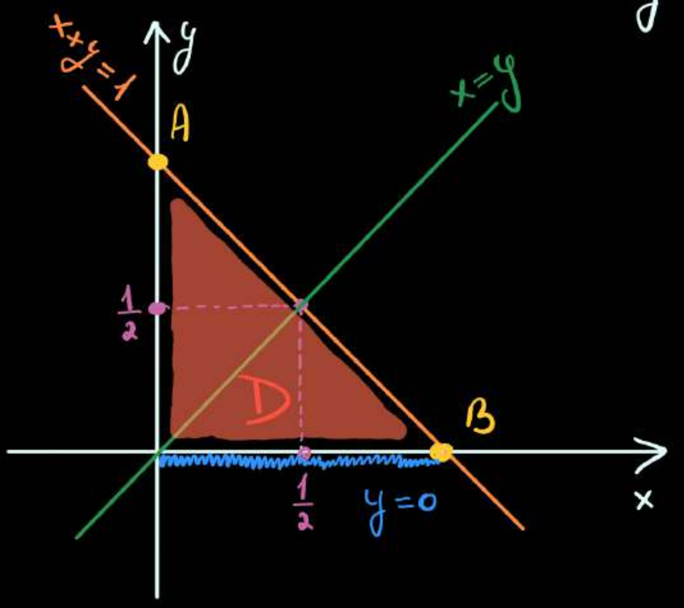
e. $E(X \cdot Y | B) = ?$ $\quad Var(X \cdot Y | B) = ?$

f. marginal PDFs, $f_X(x) = ?$ $f_Y(y) = ?$

g. $cov(X, Y) = ?$

h. X, Y - independent?

a. $$\boxed{f_{X,Y} \text{ - PDF} \iff \iint_{\mathbb{R}^2} f_{X,Y}(x,y)\, dx\, dy = 1} \Rightarrow$$



$x + y = 1$: $\bullet\ x = 0 \Rightarrow y = 1 \Rightarrow \underline{A(0;1)}$

$\bullet\ y = 0 \Rightarrow x = 1 \Rightarrow \underline{B(1;0)}$

$$\Rightarrow \iint_D f_{X,Y}(x,y)\, dx dy = \iint_D c\, dx dy \Rightarrow$$

$$\text{proj}_{Ox} D = [0,1] \Rightarrow \underline{x \in [0,1]}$$

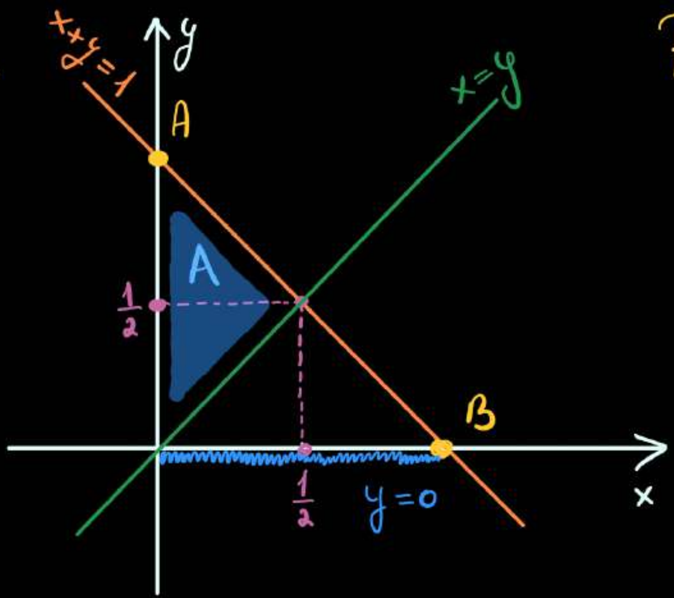
$$y \in [\underset{\substack{\downarrow \\ y=0}}{OB}, \underset{\substack{\downarrow \\ x+y=1 \Rightarrow y = 1-x}}{AB}] \Rightarrow y \in \underline{[0, 1-x]}$$

$$\Rightarrow \int_0^1 dx \int_0^{1-x} c\, dy = c \int_0^1 dx \int_0^{1-x} dy = c \cdot \int_0^1 (1-x)\,dx = c\left(x\Big|_0^1 - \frac{x^2}{2}\Big|_0^1 \right) =$$

$$= c\left(1 - \frac{1}{2}\right) = c \cdot \frac{1}{2} = 1 \Rightarrow \boxed{c = 2} \Rightarrow$$

$$\Rightarrow f_{X,Y}(x,y) = \begin{cases} 2, & x+y \leq 1,\ x \geq 0,\ y \geq 0 \\ \text{zero}, & \text{otherwise} \end{cases}$$

b.

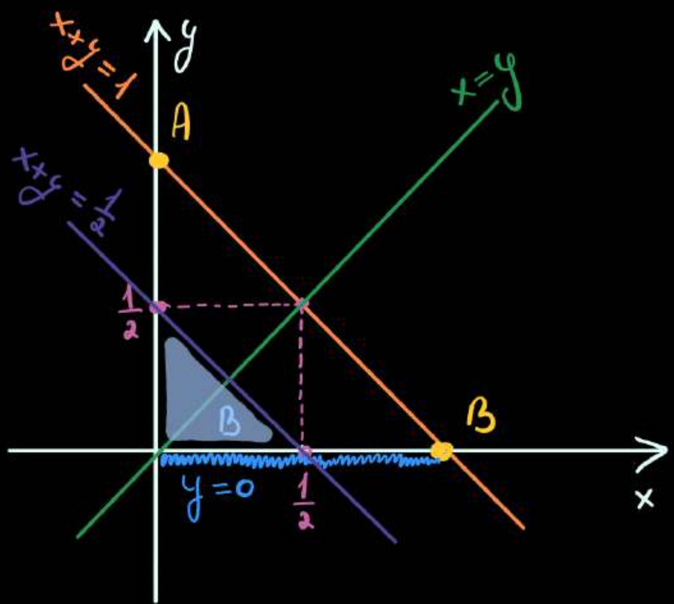


$$P(X \leq Y) = \iint_A f_{X,Y}(x,y)\,dx\,dy =$$

$$= \iint_A 2\,dx\,dy = 2 \cdot \iint_A dx\,dy = 2 \cdot A_A$$

$$= 2 \cdot \underbrace{\frac{1}{2} \cdot A_D}_{= A_A} = 2 \cdot \frac{1}{2} \cdot \frac{1}{2} = 2$$

c.



$$x+y = \frac{1}{2} : \ \bullet\, x = 0 \Rightarrow y = \frac{1}{2}$$

$$\bullet\, y = 0 \Rightarrow x = \frac{1}{2}$$

$$P(X+Y \leq 2) = P(B) =$$

$$= \iint_B 2 \cdot dx\,dy = 2 \cdot \iint_B dx\,dy =$$

$$= 2 \cdot A_B = \cancel{2} \cdot \frac{\frac{1}{2} \cdot \frac{1}{2}}{\cancel{2}} = \frac{1}{4}$$

d. $f_{X,Y|A}(x,y) = \begin{cases} \dfrac{f_{X,Y}(x,y)}{P(A)}, & (x,y) \in A \\ \text{zero, otherwise} \end{cases}$

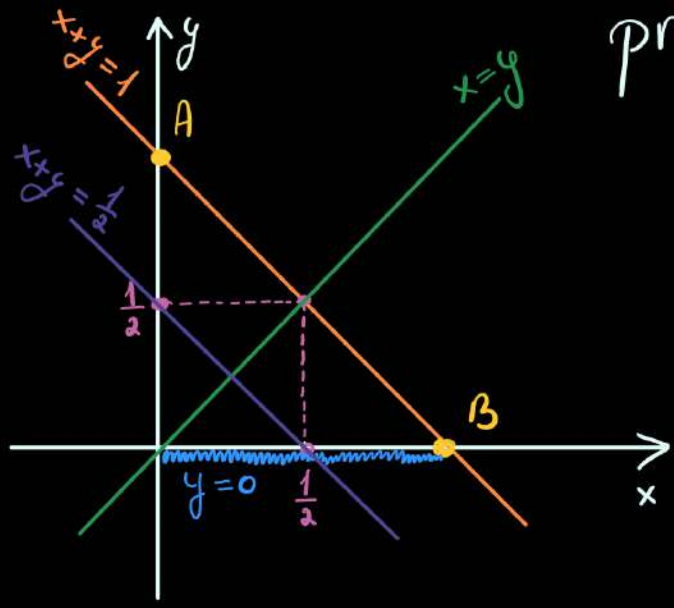
$$\boxed{f_{X,Y|A}(x,y) = \begin{cases} \dfrac{2}{\frac{1}{2}} = 4, & x \geq 0, y \geq 0, x+y \leq 1, x \leq y \\ \text{zero, otherwise} \end{cases}}$$

$$f_{X,Y|B}(x,y) = \begin{cases} \dfrac{f_{X,Y}(x,y)}{P(B)}, & (x,y) \in A \\ \text{zero, otherwise} \end{cases}$$

$$\boxed{f_{X,Y|B}(x,y) = \begin{cases} \dfrac{2}{\frac{1}{4}} = 8, & x \geq 0, y \geq 0, x+y \leq \frac{1}{2} \\ \text{zero, otherwise} \end{cases}}$$

e. $\boxed{E(g(X,Y)|B) = \iint_{\mathbb{R}^2} g(x,y) \cdot \underline{f_{X,Y|B}(x,y)}\, dx\, dy} \Rightarrow$

$$\Rightarrow E(X \cdot Y|B) = \iint_{\mathbb{R}^2} x \cdot y \cdot f_{X,Y|B}(x,y)\, dx\, dy = \iint_{B} x \cdot y \cdot 8\, dx\, dy = 8 \iint_{B} dx\, dy \Rightarrow$$



$$\text{proj}_{Ox} B = \left[0, \tfrac{1}{2}\right] \Rightarrow$$

$$\Rightarrow \underline{x \in \left[0, \tfrac{1}{2}\right]}$$

$$\Rightarrow \underline{y \in \left[0, \tfrac{1}{2} - x\right]}$$

$$\Rightarrow 8 \int_0^{\frac{1}{2}} x \, dx \int_0^{\frac{1}{2}-x} y \, dy = 8 \int_0^{\frac{1}{2}} x \left(\frac{y^2}{2}\right)\Big|_0^{\frac{1}{2}-x} dx$$

$$= \overset{4}{\cancel{8}} \int_0^{\frac{1}{2}} x \, \frac{\left(\frac{1}{4} + x^2 - x\right)}{\cancel{2}} \, dx =$$

$$= \left(\frac{x^2}{2} + \frac{x^4}{16} - \frac{x^3}{12}\right)\Big|_0^{\frac{1}{2}} = \frac{1}{8} - \frac{1}{6} + \frac{1}{16} = \underline{\frac{1}{48}}$$

$$\boxed{Var(X \cdot Y | B) = E\left((X \cdot Y)^2 | B\right) - \left[E(X \cdot Y | B)\right]^2}$$

$$E\left((X \cdot Y)^2 | B\right) = \iint_{\mathbb{R}^2} (x \cdot y)^2 \cdot \underline{f_{X,Y|B}}(x,y) \, dx \, dy =$$

$$= \iint_B x^2 \cdot y^2 \cdot 8 \, dx \, dy = 8 \int_0^{\frac{1}{2}} x^2 \, dx \int_0^{\frac{1}{2}-x} y^2 \, dy =$$

$$= 8 \int_0^{\frac{1}{2}} x^2 \left(\frac{y^3}{3}\right)\Big|_0^{\frac{1}{2}-x} dx = \frac{8}{3} \int_0^{\frac{1}{2}} x^2 \left(\frac{1}{2} - x\right)^3 dx =$$

$$= \frac{8}{3} \int_0^{\frac{1}{2}} x^2 \left(-x^3 + \frac{3}{2}x^2 - \frac{3}{4}x + \frac{1}{8}\right) dx = \ldots$$

f. $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy = \int_0^{1-x} 2\,dy = 2\cdot y \Big|_0^{1-x} = 2(1-x)$

$$\Rightarrow \boxed{f_X(x) = 2-2x}$$

$$\Rightarrow f_X(x) = \begin{cases} 2-2x, & x\in[0,1] \\ \text{zero}, & \text{otherwise} \end{cases}$$

Because we have a symmetrical domain and a symmetrical function $\Rightarrow$

$$f_Y(y) = \begin{cases} 2-2y, & y\in[0,1] \\ \text{zero}, & \text{otherwise} \end{cases}$$

g. $\boxed{\mathrm{cov}(X,Y) = E(X\cdot Y) - E(X)\cdot E(Y)}$

$$E(X,Y) = \iint_{\mathbb{R}^2} x\cdot y\cdot f_{X,Y}(x,y)\,dx\,dy = \int_0^1 dx \int_0^{1-x} x\cdot y\cdot 2\,dy =$$

$$= 2\int_0^1 x\,dx \int_0^{1-x} y\,dy = \cancel{2}\int_0^1 x\,\frac{(1-x^2)}{\cancel{2}}\,dx = \int_0^1 x\,(1-2x+x^2)\,dx =$$

$$= \int_0^1 x-2x^2+x^3\,dx = \left(\frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4}\right)\Big|_0^1 = \frac{1}{2} - \frac{2}{3} + \frac{1}{4} = \frac{1}{12}$$

$$E(X) = \iint_{\mathbb{R}^2} x \cdot f_{X,Y}(x,y)\,dx\,dy = \int_0^1 dx \int_0^{1-x} 2\,dy = 2\int_0^1 dx \int_0^{1-x} 1\,dy =$$

$$= 2\int_0^1 x\cdot(1-x)\,dx = 2\int_0^1 x - x^2\,dx = 2\left(\frac{x^2}{2} - \frac{x^3}{3}\right)\Big|_0^1 = 2\left(\frac{1}{2} - \frac{1}{3}\right)$$

$$\Rightarrow E(X) = \frac{1}{3}$$

Because we have a symmetrical domain and a symmetrical function $\Rightarrow E(Y) = E(X) = \frac{1}{3}$

$$\Rightarrow cov(X,Y) = \frac{1}{12} - \frac{1}{3}\cdot\frac{1}{3} = \frac{1}{12} - \frac{1}{9} = \frac{3}{36} - \frac{4}{36} = -\frac{1}{36}$$

$\Rightarrow X, Y$ are not INDEPENDENT

$$\boxed{X, Y - \text{independent} \Leftrightarrow f_{X,Y}(x,y) = f_X(x)\cdot f_Y(y)}$$

Homework: X, Y - r.v. have the joint:

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{2}, & -1 \le x \le y \le 1 \\ \text{zero, otherwise} \end{cases}$$

a. $P(X>0) = ?$ b. $f_X(x) = ?$ c. $E(X) = ?$ d. $E(e^{X+Y}) = ?$